

BANGKOK DON MUEANG INTL, Thailand

WMO: 484560

Lat: 13.913N

Lon: 100.607E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.5	21.0	11.1	8.2	25.1	13.0	9.3	25.7	6.4	27.5	5.9	27.8	2.3	20	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.9	37.2	26.5	36.4	26.6	35.8	26.5	29.6	33.8	29.0	33.2	28.6	32.5	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.6	25.1	32.0	28.1	24.2	31.4	27.6	23.6	30.9	97.5	33.6	94.8	33.5	92.5	32.4	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.4	6.4	5.7	DB	17.0	38.5	2.1	1.1	15.5	39.3	14.2	40.0	13.1	40.6	11.5	41.4	
			WB	13.8	30.0	2.2	1.3	12.2	31.0	10.9	31.8	9.7	32.5	8.1	33.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	27.4	28.9	30.3	31.2	30.7	30.1	29.7	29.6	29.2	29.0	28.5	27.2
	DBStd	2.00	1.94	1.77	1.74	1.66	1.69	1.24	1.25	1.18	1.15	1.32	1.76	2.30
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7054	539	529	630	637	641	604	612	608	576	590	554	534
	CDD18.3	4013	281	296	372	387	383	354	353	350	326	332	304	276
	CDH23.3	49672	3065	3582	4862	5316	5082	4607	4447	4342	3835	3901	3589	3044
	CDH26.7	23738	1277	1685	2551	3007	2706	2306	2092	2005	1658	1652	1542	1257
(14) Wind	WSAvg	2.7	2.2	2.6	3.1	3.1	2.9	3.1	3.3	3.2	2.7	2.1	2.2	2.3
(15) Precipitation	PrecAvg	1503	18	15	50	88	209	186	177	200	313	236	41	9
	PrecMax	2039	111	57	163	250	454	353	307	370	494	443	126	44
	PrecMin	1011	0	0	2	6	75	47	41	71	199	58	0	0
	PrecStd	243	25	16	47	69	98	76	71	83	73	83	38	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.8	36.3	37.6	38.6	38.2	37.1	36.0	35.8	35.2	35.2	35.2	34.8
		MCWB	24.3	25.5	26.2	26.2	26.4	25.9	27.0	27.6	27.8	26.9	25.6	24.6
	2%	DB	33.9	35.1	36.2	37.2	36.9	35.5	34.9	34.8	34.2	34.1	34.1	33.6
		MCWB	24.2	25.2	25.9	26.6	26.9	26.8	26.7	26.9	27.0	26.3	25.4	24.3
	5%	DB	32.9	34.2	35.6	36.5	35.9	34.6	34.0	33.9	33.5	33.1	33.2	32.8
		MCWB	23.8	25.1	25.9	26.7	26.8	26.8	26.5	26.7	26.7	26.1	25.3	24.0
	10%	DB	32.0	33.3	34.8	35.8	34.8	33.8	33.1	33.0	32.8	32.2	32.2	31.9
		MCWB	23.6	24.8	25.9	26.8	26.8	26.7	26.3	26.3	26.5	25.9	25.0	23.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	28.3	29.3	30.2	29.7	29.8	29.4	29.2	30.2	29.8	28.3	27.0
		MCDB	30.9	33.2	34.7	35.1	34.1	34.0	33.5	33.1	33.7	33.0	31.6	30.3
	2%	WB	26.6	27.4	28.5	29.4	29.0	29.0	28.6	28.4	28.9	28.4	27.2	26.2
		MCDB	30.4	32.1	33.3	34.1	33.4	33.0	32.3	32.2	32.0	31.0	30.8	30.2
	5%	WB	26.0	26.7	27.7	28.7	28.3	28.2	28.0	27.7	28.1	27.6	26.7	25.6
		MCDB	29.5	31.1	32.1	33.1	32.4	31.6	31.3	31.4	31.2	30.4	30.4	29.9
	10%	WB	25.4	26.2	27.1	28.1	27.8	27.6	27.4	27.2	27.3	26.9	26.2	25.0
		MCDB	29.1	30.1	31.4	32.1	31.8	31.0	30.8	30.9	30.6	30.1	29.9	29.6
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.2	8.9	8.9	8.1	7.6	7.2	7.2	7.4	7.1	7.5	8.5
		MCDBR	9.9	9.8	10.0	9.6	9.1	8.6	8.1	8.1	8.4	7.9	8.2	8.9
		MCWBR	3.3	3.3	3.4	3.3	3.1	3.2	3.1	3.0	3.5	3.0	3.0	2.8
	5% WB	MCDBR	8.5	9.1	9.2	9.1	8.2	7.8	7.5	7.6	8.0	7.4	7.5	7.7
		MCWBR	3.3	3.4	3.6	4.0	3.7	3.7	3.5	3.5	4.3	3.6	3.2	2.9
(40) Clear-Sky Solar Irradiance	taub		0.544	0.615	0.645	0.583	0.504	0.467	0.459	0.460	0.486	0.578	0.522	0.509
	taud		1.954	1.811	1.772	1.959	2.236	2.336	2.361	2.344	2.261	1.962	2.096	2.076
	Ebn at Noon		756	721	712	756	805	826	835	843	826	740	771	775
	Edh at Noon		181	217	231	191	142	127	125	128	139	184	156	157
(44) All-Sky Solar Radiation	RadAvg		5.04	5.46	5.63	6.05	5.55	5.29	5.04	5.07	4.91	4.80	4.93	4.88
	RadStd		0.29	0.23	0.41	0.36	0.39	0.28	0.25	0.32	0.27	0.27	0.32	0.23

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.23	+0.25	N/A	N/A	N/A	N/A

Nomenclature: See separate page